Quality Management and Six Sigma Applications
AE310

Unit 5 & 6: Reliability Analysis

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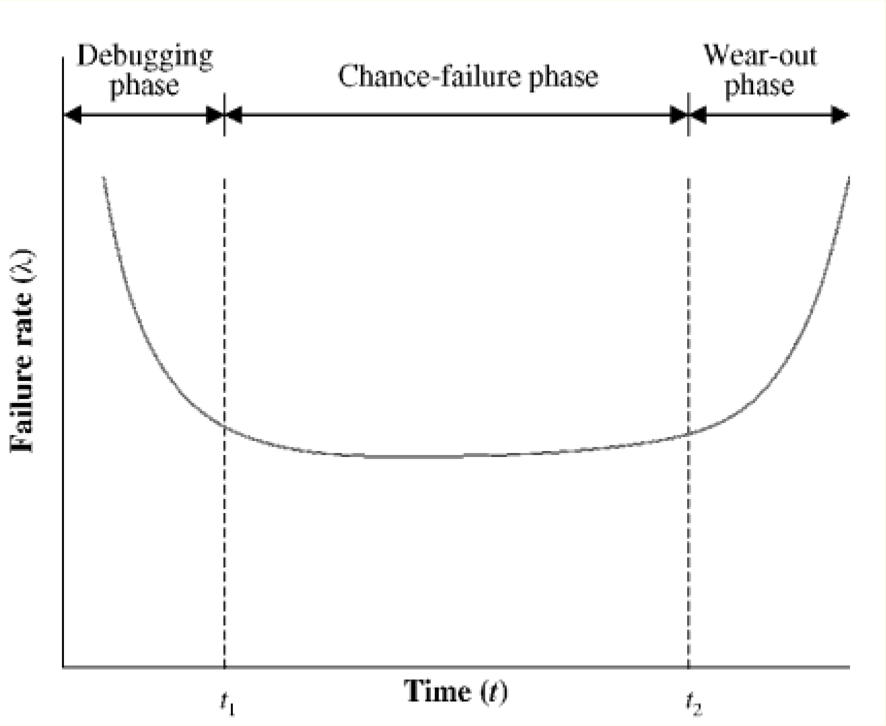
Reliability

- Reliability is the **probability** of a product performing its **intended function** for a **stated period of time** under certain **specified conditions**.
- Four aspects of reliability are apparent from this definition:
  1. Reliability is a probability-related concept.
  2. The functional performance of the product has to meet certain stipulations.
  3. Reliability implies successful operation over a certain period of time.
  4. Operating or environmental conditions under which product use takes place are specified.
- Example: *The reliability of a nylon cord is 0.92 in bearing loads of 1000 kg for two years under dry conditions.*

Life-cycle Curve

- Most products go through **three distinct phases from product inception to wear-out**.
- This curve is often referred to as the **bathtub curve**; it consists of the **debugging phase**, the **chance-failure phase**, and the **wear-out phase**.
- The **debugging phase** exhibits a drop in the failure rate as initial problems identified during prototype testing are ironed out.
- The **chance failure phase**, between t_1 and t_2 times and, is then encountered; failures occur randomly and independently. This phase, in which the failure rate is constant, typically represents the **useful life of the product**.
- Following this is the **wear-out phase**, in which an increase in the failure rate is observed. Here, at the end of their useful life, parts age and wear out



Probability Distributions to Model Failure Rate

- For the **chance-failure phase**, which represents the useful life of the product, **the failure rate is constant**. As a result, the **exponential distribution** can be used to describe the time to failure of the product for this phase.

$$f(t) = \lambda e^{-\lambda t}, \qquad t \geq 0$$

Where λ = failure rate and t = time.

- If failure rate is constant, **mean time to failure (MTTF)** = 1/ λ
- For **repairable equipment**, **mean time between failure (MTBF)** = 1/λ
- The reliability at time *t*, *R*(*t*), is the probability of the product lasting up to at least time *t*. It is given by

$$R(t) = 1 - F(t) = 1 - \int_0^t e^{-\lambda t}\,dt = e^{-\lambda t}$$

Reliability Analysis – Problems

Problem 1

An amplifier has an exponential time-to-failure distribution with a failure rate of 8% per 1000 hours. What is the reliability of the amplifier at 5000 hours? Find the mean time to failure.

Reliability Analysis – Problems

Problem 2

What is the highest failure rate for a product if it is to have a probability of survival (i.e., successful operation) of 95% at 4000 hours? Assume that the time to failure follows an exponential distribution.

Availability

- The availability of a system at time t is the **probability that the system will be up and running** at time t.
- To improve availability, maintenance procedures are incorporated, which may include periodic or preventive maintenance or condition-based maintenance.
- An availability index is defined as

$$\text{Availability} = \frac{\text{operating time}}{\text{operating time} + \text{downtime}}$$

Availability

- For a steady-state system, denoting the mean time to repair (MTTR) to include all the various components of down- time, we have

$$\text{Availability} = \frac{\text{MTTF}}{\text{MTTF} + \text{MTTR}}$$

- In the situation when the time-to-failure distribution is exponential (with a failure rate λ) and the time-to-repair distribution is also exponential (with a repair rate μ), the availability is given by $\mu/(\lambda + \mu)$.

Reliability Analysis – Problems

Problem 3

A manufacturing unit operates a milling machine. On average, the machine runs for 400 hours before it fails. Once it fails, the maintenance team takes about 20 hours to repair and restart it. Determine MTTF, MTTR, and availability.

Reliability Analysis – Problems

Problem 4

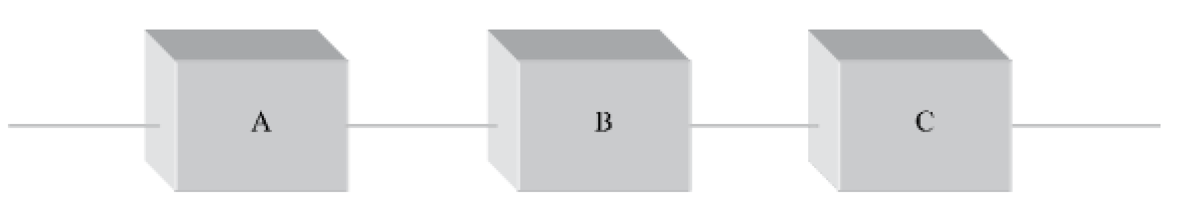
A hospital MRI machine is operational 90% of the time. On average, it works for 180 hours before failure. Determine failure rate, repair rate, and MTTR.

System Reliability

- Most products are made up of a number of components.
- The reliability of each component and the configuration of the system consisting of these components determine the **system reliability** (i.e., the reliability of the product).
- Although product design, manufacture, and maintenance influence reliability, improving reliability is largely the domain of design.
- One common approach for increasing the reliability of the system is through **redundance in design**, which is usually achieved by placing components in parallel: As long as one component operates, the system operates.

Systems with Components in Series

- Figure shows **a system with three components** (A, B, and C**) in series**. For the system to operate, each component must operate. It is assumed that the components operate independent of each other (i.e., the failure of one component has no influence on the failure of any other component).



- In general, if there are n components in series, where the reliability of the *i*th component is denoted by R_i, the system reliability R_s is

$$\boldsymbol{R_s = R_1 \times R_2 \times \ldots\ldots \times R_i \times \ldots\ldots \times R_n} \quad \text{and} \quad \mathbf{MTTF} = \boldsymbol{1/n\lambda}$$

- The system reliability decreases as the number of components in series increases.

Reliability Analysis – Problems

Problem 5

A module of a satellite monitoring system has 500 components in series. The reliability of each component is 0.999. Find the reliability of the module. If the number of components in series is reduced to 200, what is the reliability of the module?

Reliability Analysis – Problems

Problem 6

The automatic focus unit of a television camera has 10 components in series. Each component has an exponential time-to-failure distribution with a constant failure rate of 0.05 per 4000 hours. What is the reliability of each component after 2000 hours of operation? Find the reliability of the automatic focus unit for 2000 hours of operation. What is its mean time to failure?

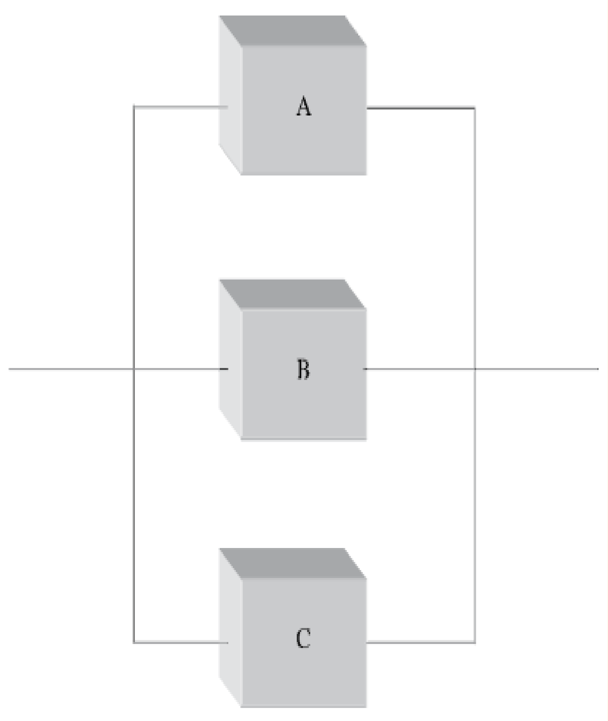
Systems with Components in Parallel

- Figure shows **a system with three components** (A, B, and C**) in parallel**. The components are redundant; the system operates as long as at least one of the components operates. The only time the system fails is when all the parallel components fail.
- In general, if there are n components in parallel, where the reliability of the *i*th component is denoted by R_i, the system reliability R_s is

$$R_s = 1 - (1 - R_1)\times(1 - R_2)\times\ldots\ldots\times(1 - R_i)\times\ldots\ldots\times(1 - R_n) \text{ and}$$

$$MTTF = (1/\lambda)\times(1+1/2+1/3+\ldots..+1/n)$$

- The system reliability increases as the number of components in parallel decreases.



Reliability Analysis – Problems

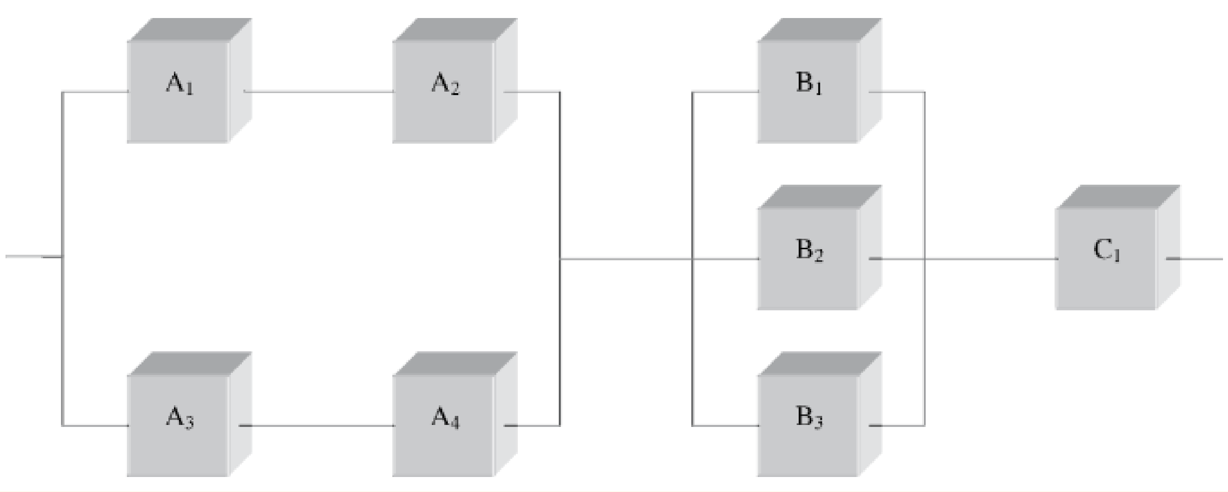
Problem 7

Consider a system with three parallel components, determine the system reliability for 2000 hours of operation and find the mean time to failure. Assume that all three components have an identical time-to-failure distribution that is exponential, with a constant failure rate of 0.0005/hour. What is the mean time to failure of each component? If it is desired for the system to have a mean time to failure of 4000 hours, what should the mean time to failure be for each component?

Reliability Analysis – Problems

Problem 8

Find the reliability of the eight-component system shown in the figure; some components are in series and some are in parallel. The reliabilities of the components are as follows: $R_{A1} = 0.92$, $R_{A2} = 0.90$, $R_{A3} = 0.88$, $R_{A4} = 0.96$, $R_{B1} = 0.95$, $R_{B2} = 0.90$, $R_{B3} = 0.92$, and $R_{C1} = 0.93$.



Reliability of k-out-of-n System

- For the successful operation of a system having **n** components:
  - ✓ it is necessary that **all n components** must perform their intended function successfully in the case of **series system**, and
  - ✓ only **one component** needs to work successfully in the case of **parallel system**.
- However, in practice, there exist systems, which require the successful operation of **at least k components** ($1 \leq k \leq n$), out of **n identical components**.
- Such systems are known as **k-out-of-n systems** or **partially redundant systems**.

Reliability of k-out-of-n System

Redundancy

- Redundancy in a system means that **there exists an alternative parallel path for the successful operation of the system**.
- For example, a system has **n** components connected in **parallel** and at least **k** ($1 \le k \le n$) components are needed for the successful operation of the system. In such a system, we say that **the number of basic components is k** and the remaining **(n – k) components are known as redundant components**.

Reliability of k-out-of-n System

Fully Redundant System

A k-out-of-n system is known as a fully redundant system if **k = 1**. But for k = 1, the k-out-of-n system is the same as the **parallel system**. Thus, the parallel system is nothing but a **1-out-of-n system** and is also known as a **fully redundant system**.

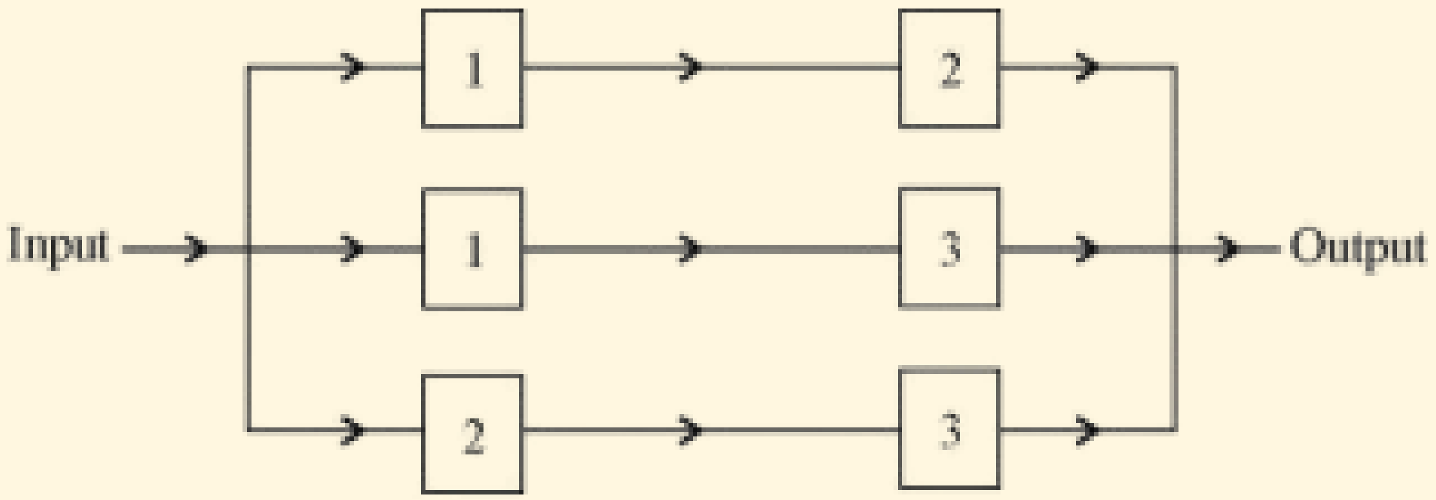
Non-redundant System

A k-out-of-n system is said to be a non-redundant system if **k = n**. But for k = n, the k-out-of-n system is simply **the series system**. Thus, a series system is nothing but an **n-out-of-n system** and is also known as **non-redundant system**.

Reliability of k-out-of-n System

Example

2-out-of-3 system



Reliability of k-out-of-n System

- Consider a k-out-of-n system having **n identical and independent components**. Successful operation a component in such system is considered a success. Thus, reliability of each component represents the probability of success. Since there are **n** components, the number of successes for a given time may be 0, 1, 2, …, n.
- Applying binomial distribution, the reliability of a k-out-of-n system (***R_s***) for a given time can be calculated as

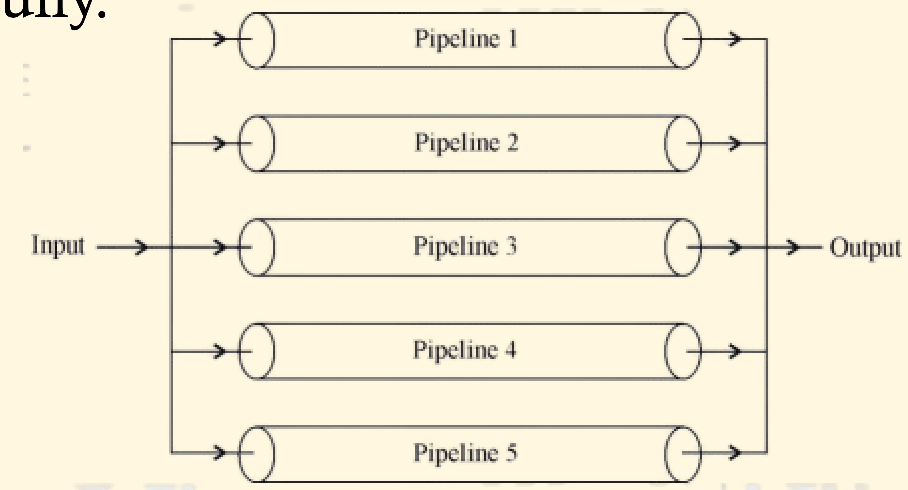
$$R = \sum_{x=k}^{n} \binom{n}{x} R^x (1 - R)^{n-x}$$

where, R = reliability of each component

Reliability Analysis – Problems

Problem 9

Consider a piping system having 5 pipes connected in parallel as shown in the figure. Assume that all pipes are identical and independent. If reliability of smooth flow of the liquid from each pipeline is 0.80 for a mission of 1 year, calculate the reliability of the system working successfully. The system is said to work successfully if at least 3 pipelines perform their intended function successfully.



Reliability Analysis – Problems

Problem 10

Consider a piping system having 4 pipes connected in parallel as shown in the figure. Assume that all pipes are independent but not identical. It is given that the reliabilities of 4 pipelines for a mission of 1000 hours are $R_1 = 0.60$, $R_2 = 0.70$, $R_3 = 0.80$ and $R_4 = 0.75$, respectively. Determine the reliability of 2-out-of-4 system.